Number Theory — problem 6.147

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Let $a, b, c \in \mathbb{N}$ with $\gcd(a, b) = \gcd(b, c) = \gcd(c, a) = 1$. Prove that $2abc - ab - bc - ca$ is the largest integer which cannot be expressed in the form $xbc + yca + zab$, where x, y, z are nonnegative integers (IMO 1983).